Student Performance Analytics

HiveQL Step-by-Step Query Execution & Viva Defense Guide

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1. Infrastructure Setup & Dual Execution Architecture

This runbook guides you through executing all **17 Analytical HiveQL Queries** across 7 modular scripts. Our project features a **Dual-Engine Architecture**: queries can be executed directly inside the distributed **Apache Hadoop/Hive Docker Cluster** or via the **Automated Python Query Runner**.

Step	Target Environment	Exact Terminal Command	Verification Check
1. Start Cluster	Docker Containers	`docker compose -f docker/docker-compose.yml up -d`	NameNode (9870), DataNode (9864), Hive (10000) active
2. HDFS Ingest	Hadoop Distributed FS	`./docker/init-hadoop.sh`	File at /user/hadoop/student-performance/student-por.csv
3. Run All Queries	Automated Pipeline	`./scripts/run_pipeline.sh`	Executes 17 queries & writes CSVs to output/query-results/
4. Launch UI	Streamlit Dashboard	`streamlit run app.py`	Interactive UI at http://localhost:8501 (Live HDFS Connected)

2. DDL & Storage Ingestion Scripts (01 to 03)

Script 01: Database Creation (hive/01_create_database.sql)

```
CREATE DATABASE IF NOT EXISTS student_performance_bda;
USE student_performance_bda;
```

Run Command: docker exec -i student_bda_hiveserver hive -f /hive/01_create_database.sql
Technical Rationale: Creates the isolated warehouse namespace in the Hive Metastore.

Script 02: External Table Schema (hive/02_create_table.sql)

```
CREATE EXTERNAL TABLE IF NOT EXISTS student_performance (
school STRING, sex STRING, age INT, address STRING, famsize STRING, Pstatus STRING,
Medu INT, Fedu INT, Mjob STRING, Fjob STRING, reason STRING, guardian STRING,
traveltime INT, studytime INT, failures INT, schoolsup STRING, famsup STRING,
paid STRING, activities STRING, nursery STRING, higher STRING, internet STRING,
romantic STRING, famrel INT, freetime INT, goout INT, Dalc INT, Walc INT,
health INT, absences INT, G1 INT, G2 INT, G3 INT
)
ROW FORMAT DELIMITED FIELDS TERMINATED BY ';'
STORED AS TEXTFILE LOCATION '/user/hadoop/student-performance/';
```

Run Command: docker exec -i student_bda_hiveserver hive -f /hive/02_create_table.sql
Technical Rationale: Schema-on-Read: Semicolon delimited (;) parsing across 33 student attributes.

Script 03: Data Ingestion (hive/03_load_data.sql)

```
LOAD DATA INPATH '/user/hadoop/student-performance/student-por.csv'
INTO TABLE student_performance;
```

Run Command: docker exec -i student_bda_hiveserver hive -f /hive/03_load_data.sql
Technical Rationale: Mounts HDFS block bytes into the Hive metastore table catalog.

3. Step-by-Step HiveQL Analytical Queries (1 to 17)

Query 1: Total Student Enrollment Count

File: hive/04_basic_analysis.sql

```
SELECT COUNT(*) AS total_students FROM student_performance;
```

Expected Output:
total_students: 649

Key Insight: *Ingests exactly 649 student records from the UCI Portuguese dataset without loss.*

Teacher Viva Defense: Proves data integrity during HDFS block allocation and table loading.

Query 2: Final Grade (G3) Summary Statistics

File: hive/04_basic_analysis.sql

```
SELECT ROUND(AVG(G3), 2) AS avg_final, MIN(G3) AS min_final,
MAX(G3) AS max_final, ROUND(STDDEV(G3), 2) AS std_final
FROM student_performance;
```

Expected Output:
avg_final: 11.91 | min_final: 0 | max_final: 19 | std_final: 3.23

Key Insight: *Average cohort grade is 11.91/20 with a standard deviation of 3.23 points.*

Teacher Viva Defense: Demonstrates aggregation functions (AVG, MIN, MAX, STDDEV) over distributed records.

Query 3: Period-over-Period Progression Tracking (G1 -> G2 -> G3)

File: hive/04_basic_analysis.sql

```
SELECT ROUND(AVG(G1), 2) AS avg_G1, ROUND(AVG(G2), 2) AS avg_G2,
ROUND(AVG(G3), 2) AS avg_G3 FROM student_performance;
```

Expected Output:
avg_G1: 11.40 | avg_G2: 11.57 | avg_G3: 11.91

Key Insight: *Positive learning curve: cohort average improved by +0.51 points across the academic year.*

Teacher Viva Defense: Proves student retention and gradual knowledge accumulation across periods.

Query 4: Absence Distribution Summary

File: hive/04_basic_analysis.sql

```
SELECT ROUND(AVG(absences), 2) AS avg_abs, MIN(absences) AS
min_abs,
MAX(absences) AS max_abs, ROUND(STDDEV(absences), 2) AS std_abs
FROM student_performance;
```

Expected Output:
avg_abs: 3.66 | min_abs: 0 | max_abs: 32 | std_abs: 4.64

Key Insight: *Mean absences is 3.66 days; high variance up to 32 days reveals severe attendance outliers.*

Teacher Viva Defense: Highlights attendance variance as a critical behavioral risk variable.

Query 5: Academic Performance Disaggregated by Gender

File: hive/05_performance_analysis.sql

```
SELECT sex, COUNT(*) AS count, ROUND(AVG(G1), 2) AS avg_G1,
ROUND(AVG(G2), 2) AS avg_G2, ROUND(AVG(G3), 2) AS avg_G3,
ROUND(AVG(absences), 2) AS avg_abs
FROM student_performance GROUP BY sex;
```

Expected Output:
Female (383): avg_G3=12.25, abs=3.58 | Male (266): avg_G3=11.41, abs=3.78

Key Insight: *Female students outperformed male students by +0.84 points with slightly better attendance.*

Teacher Viva Defense: Showcases multi-column GROUP BY aggregation and gender equity analysis.

Query 6: Institutional Comparison: Gabriel Pereira (GP) vs Mousinho (MS)

File: hive/05_performance_analysis.sql

```
SELECT school, COUNT(*) AS students, ROUND(AVG(G3), 2) AS avg_G3,
ROUND(AVG(studytime), 2) AS avg_study, ROUND(AVG(absences), 2) AS
avg_abs
FROM student_performance GROUP BY school;
```

Expected Output:
GP (423): avg_G3=12.58, study=2.01h, abs=4.22 | MS (226): avg_G3=10.65, study=1.77h, abs=2.62

Key Insight: *Urban school GP leads rural MS by +1.93 grade points due to higher average study time.*

Teacher Viva Defense: Identifies geographic/resource disparities between urban and rural school systems.

Query 7: Cohort Pass vs Fail Ratio (Passing Threshold G3 >= 10)

File: hive/05_performance_analysis.sql

```
SELECT CASE WHEN G3 >= 10 THEN 'Passed' ELSE 'Failed' END AS
status,
COUNT(*) AS student_count,
ROUND(COUNT(*) * 100.0 / 649, 2) AS percentage
FROM student_performance GROUP BY CASE WHEN G3 >= 10 THEN 'Passed'
ELSE 'Failed' END;
```

Expected Output:
Passed: 549 students (84.59%) | Failed: 100 students (15.41%)

Key Insight: *Overall graduation pass rate is strong at 84.6%, but 15.4% require urgent intervention.*

Teacher Viva Defense: Uses conditional CASE-WHEN logic inside GROUP BY for executive KPI classification.

Query 8: Grade Distribution by Academic Tiers

File: hive/05_performance_analysis.sql

```
SELECT CASE
WHEN G3 < 10 THEN '01. Fail (0-9)'
WHEN G3 BETWEEN 10 AND 13 THEN '02. Sufficient (10-13)'
WHEN G3 BETWEEN 14 AND 15 THEN '03. Good (14-15)'
ELSE '04. Excellent (16-20)'
END AS grade_category, COUNT(*) AS count,
ROUND(COUNT(*) * 100.0 / 649, 2) AS pct,
ROUND(AVG(studytime), 2) AS avg_study, ROUND(AVG(absences), 2) AS
avg_abs
FROM student_performance GROUP BY ... ORDER BY grade_category;
```

Expected Output:
Fail: 15.4% (1.61h study) | Sufficient: 54.7% (1.89h) | Good: 17.3% (2.13h) | Excellent: 12.6% (2.24h)

Key Insight: *Direct correlation: Study time increases monotonically with each higher grade tier.*

Teacher Viva Defense: Demonstrates multi-tier discretization and academic segmentation in Hive.

Query 9: Study Time Elasticity vs Final Academic Outcome

File: hive/06_factor_analysis.sql

```
SELECT studytime, CASE studytime WHEN 1 THEN '< 2h' WHEN 2 THEN
'2-5h'
WHEN 3 THEN '5-10h' ELSE '> 10h' END AS study_desc,
COUNT(*) AS students, ROUND(AVG(G3), 2) AS avg_G3,
ROUND(AVG(failures), 2) AS avg_fails
FROM student_performance GROUP BY studytime ORDER BY studytime;
```

Expected Output:
<2h: 10.84 avg (0.33 fails) | 2-5h: 12.09 | 5-10h: 13.23 (0.10 fails) | >10h: 13.06

Key Insight: *Studying 5-10 hours/week yields +2.39 grade points compared to <2 hours/week.*

Teacher Viva Defense: Proves behavioral factors directly mitigate academic failure rates.

Query 10: Previous Failures Impact Analysis

File: hive/06_factor_analysis.sql

```
SELECT failures, COUNT(*) AS count, ROUND(AVG(G3), 2) AS avg_G3,
ROUND(AVG(absences), 2) AS avg_abs
FROM student_performance GROUP BY failures ORDER BY failures;
```

Expected Output:
0 Failures: 12.51 (549 students) | 1 Failure: 8.64 (70) | 2: 8.81 (16) | 3: 8.07 (14)

Key Insight: *CRITICAL RISK FACTOR: Having just 1 prior failure drops average score below passing (8.64/20).*

Teacher Viva Defense: Identifies the single most decisive historical predictor of academic failure.

Query 11: Absence Tiers vs Academic Performance

File: hive/06_factor_analysis.sql

```
SELECT CASE
WHEN absences = 0 THEN '00. Zero Absences'
WHEN absences BETWEEN 1 AND 4 THEN '01. Low (1-4)'
WHEN absences BETWEEN 5 AND 9 THEN '02. Moderate (5-9)'
ELSE '03. High (10+)'
END AS absence_tier, COUNT(*) AS count, ROUND(AVG(G3), 2) AS
avg_G3
FROM student_performance GROUP BY ... ORDER BY absence_tier;
```

Expected Output:
Zero: 12.04 (244) | Low (1-4): 12.07 (222) | Moderate (5-9): 11.77 (113) | High (10+): 11.13 (70)

Key Insight: *High absenteeism (10+ missed classes) reduces student performance by nearly 1 full point.*

Teacher Viva Defense: Demonstrates tier-based clustering and attendance penalty modeling.

Query 12: Parental Educational Attainment Correlation

File: hive/06_factor_analysis.sql

```
SELECT Medu AS mother_edu, Fedu AS father_edu, COUNT(*) AS count,
ROUND(AVG(G3), 2) AS avg_G3
FROM student_performance GROUP BY Medu, Fedu HAVING count >= 10
ORDER BY avg_G3 DESC;
```

Expected Output:
Both Higher Ed (4,4): 13.12/20 (95 students) | Both Primary (1,1): 11.70/20 (20 students)

Key Insight: *Students from households with higher parental education achieve up to +1.4 points higher.*

Teacher Viva Defense: Uses HAVING clause to filter statistically robust sample sizes in distributed SQL.

Query 13: School vs Family Educational Support Effectiveness

File: hive/06_factor_analysis.sql

```
SELECT schoolsup, famsup, COUNT(*) AS count, ROUND(AVG(G3), 2) AS
avg_G3
FROM student_performance GROUP BY schoolsup, famsup ORDER BY
avg_G3 DESC;
```

Expected Output:
No School / Yes Family: 12.14 (349) | Yes School / Yes Family: 11.47 (49) | Yes School / No Family: 10.79 (19)

Key Insight: *Remedial school support is targeted at struggling students; family support helps buffer scores.*

Teacher Viva Defense: Evaluates targeted intervention effects and institutional support programs.

Query 14: Higher Education Ambition & Home Internet Access Synergies

File: hive/06_factor_analysis.sql

```
SELECT higher AS wants_higher_ed, internet AS has_home_internet,
COUNT(*) AS count, ROUND(AVG(G3), 2) AS avg_G3
FROM student_performance GROUP BY higher, internet ORDER BY avg_G3
DESC;
```

Expected Output:
Yes Higher / Yes Net: 12.53 (451) | Yes Higher / No Net: 11.40 (129) | No Higher / Any: 8.80 (69)

Key Insight: *Students aiming for university score ~3.7 points higher; internet connectivity adds +1.13 points.*

Teacher Viva Defense: Proves the compounding benefit of student motivation and digital access.

Query 15: Early Intervention Radar: At-Risk Student Identification

File: hive/07_advanced_analysis.sql

```
SELECT school, sex, age, studytime, failures, absences, G1, G2,
G3,
CASE
WHEN G3 < 10 AND failures > 0 AND absences > 10 THEN 'CRITICAL
RISK'
WHEN G3 < 10 THEN 'FAILING RISK'
WHEN failures > 0 THEN 'HISTORY RISK'
ELSE 'ATTENDANCE RISK'
END AS risk_category
FROM student_performance
WHERE G3 < 10 OR failures > 0 OR absences > 10 ORDER BY G3 ASC
LIMIT 10;
```

Expected Output:
Identified 177 at-risk students across 4 actionable severity tiers.

Key Insight: *Translates big data analytics into proactive educator intervention lists before final exams.*

Teacher Viva Defense: Demonstrates complex multi-criteria filtering (WHERE OR) and rule-based risk ranking.

Query 16: Elite Achievers Profile (G3 >= 16/20)

File: hive/07_advanced_analysis.sql

```
SELECT COUNT(*) AS top_count, ROUND(COUNT(*)*100.0/649, 2) AS
cohort_pct,
ROUND(AVG(studytime), 2) AS avg_study, ROUND(AVG(absences), 2) AS
avg_abs,
ROUND(AVG(failures), 2) AS avg_fails,
ROUND(SUM(CASE WHEN higher='yes' THEN 1 ELSE 0
END)*100.0/COUNT(*), 1) AS pct_higher_ed,
ROUND(SUM(CASE WHEN internet='yes' THEN 1 ELSE 0
END)*100.0/COUNT(*), 1) AS pct_internet
FROM student_performance WHERE G3 >= 16;
```

Expected Output:
82 Elite Students (12.63%): 2.24h study, 2.59 absences, 0.01 failures, 100% Higher Ed, 85.4% Internet

Key Insight: *100% of elite achievers desire higher education and maintain near-zero failure and low absence records.*

Teacher Viva Defense: Showcases advanced feature profiling and conditional SUM aggregation in HiveQL.

Query 17: Weekend Alcohol Consumption vs Academic Trajectory

File: hive/07_advanced_analysis.sql

```
SELECT Walc AS weekend_alcohol_level, COUNT(*) AS count,
ROUND(AVG(G3), 2) AS avg_G3, ROUND(AVG(absences), 2) AS avg_abs,
ROUND(AVG(studytime), 2) AS avg_study
FROM student_performance GROUP BY Walc ORDER BY Walc ASC;
```

Expected Output:
Level 1 (Very Low): 12.36 avg, 2.89 abs, 2.12h study (247 students)
Level 5 (Very High): 10.56 avg, 5.24 abs, 1.58h study (45 students)

Key Insight: *High weekend alcohol consumption lowers final grades by -1.8 points and nearly doubles absences (5.2 vs 2.9).*

Teacher Viva Defense: Highlights lifestyle health factors that directly undermine academic performance.

4. Essential Viva Questions & Speaking Scripts

Q1: How do you run individual HiveQL queries in front of the teacher?

Answer: *'Ma'am, we have two ways: (1) Running directly inside the Hive container: docker exec -i student_bda_hiveserver hive -e "SELECT ...", or (2) Running our automated master pipeline ./scripts/run_pipeline.sh which executes all 17 queries and outputs CSV reports.'*

Q2: What is the difference between Managed and External tables in Hive?

Answer: *'In our project we created an **EXTERNAL TABLE**. With external tables, Hive manages the metadata/schema, but if we drop the table, the raw data file remains safe inside HDFS. In a managed table, dropping the table permanently deletes both metadata and HDFS data.'*

Q3: Why does HiveQL use MapReduce/Tez under the hood?

Answer: *'Hive is a SQL abstraction on top of distributed data. When we run GROUP BY or AVG queries, Hive translates declarative SQL into distributed MapReduce or Tez execution DAGs across cluster DataNodes.'*

Q4: Which query reveals the highest correlation with student failure?

Answer: *'Query 10 (Previous Failures). A student with 0 failures averages 12.51/20, but with just 1 previous failure, the average drops below the passing threshold to 8.64/20.'*